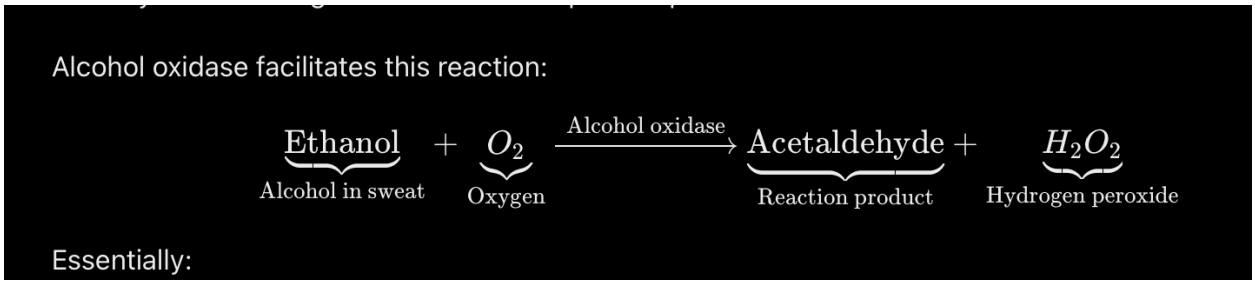
Steps to take to complete the project:

- Alcohol sensor : measures the ethanol in your sweat
- Electronic circuit : converts the sensors electrical response into usable data
- Microcontroller : processes the measurements
- Bluetooth module : sends data to the phone
- Battery : powers the wearable
- Software : estimates BAC and displays information
- Notification system : alerts the user or trusted contact

Accoriding to UC san diego research
Sensor :
Alcohol oxidase - biological component responsible for detecting alcohol
Sensor mesures hydrogen peroxide produced
The amount of hydrogen peroxide is related to the amount of ethanol prese

Things i need :

1. Pilocarpine - chemical that causes people to sweat in the spot that we need it to, this is needed because would if the person isn't sweating, how are we supposed to get them to sweat
2. Alcohol oxidase : chemical responsible for detecting alcohol



   - What is happening, why is this chemical equation important to us?

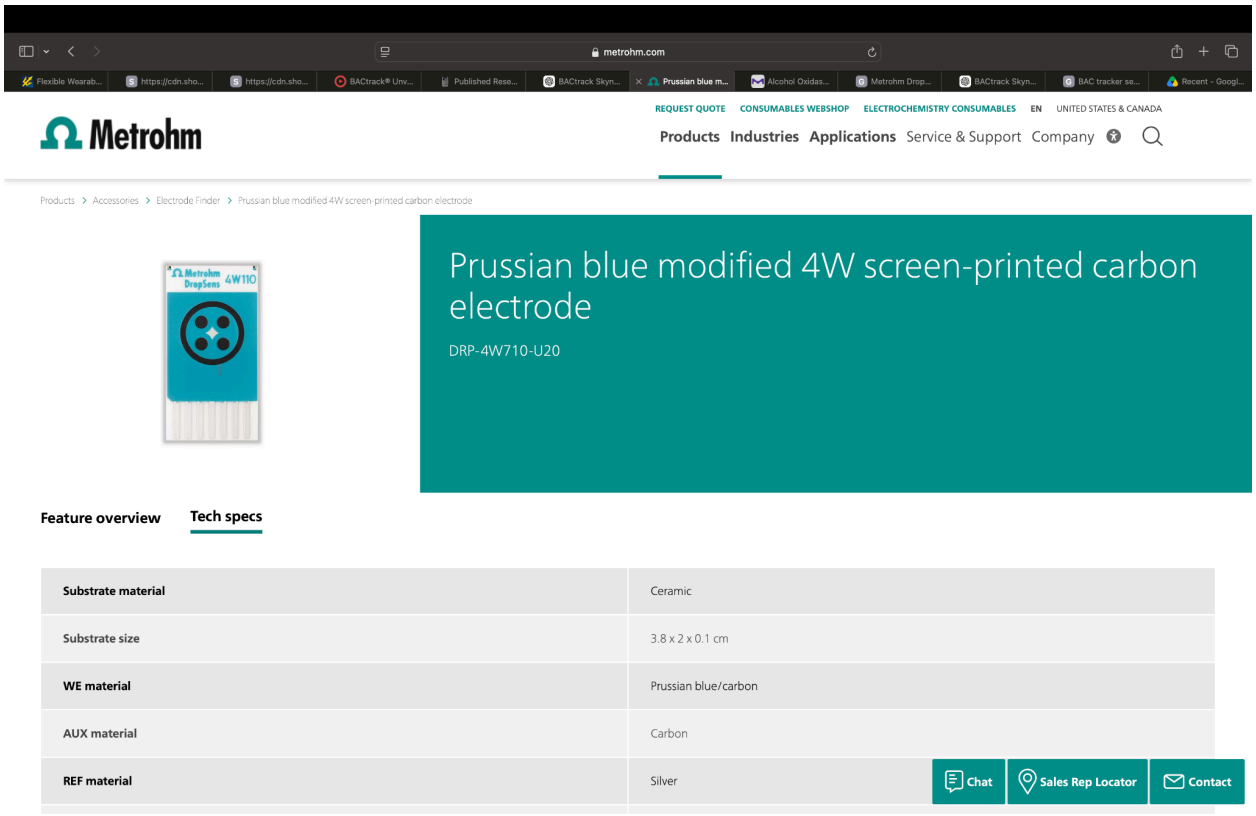
1. Ethanol in the sweat comes into contact with alcohol oxidase
2. Alchol oxidase helps ethanol react with oxygen
3. The reaction produces acetaldehyde and hydrogen peroxide
4. The electrode detects the hydrogen peroxide
5. The resulting electrical signal helps determine how much ethanol was present

So, alcohol oxidase job is to help the sensor mesure alchohl

Now, we need electrodes - contains three sensing electrodes :

- Working electrode : detects the electrochemical reaction associated with ethanol
- Reference electrode : provides a stable reference potential
- Counter electrode : completes electrical circuit

In the UC san diego research, they used :
Working electrode - prussian blue modified conductive carbon (image below)
Reference electrode - silver/silver chloride
Counter electrode - conductive carbon



The product is a small ceramic strip containing four working electrodes, one reference electrode, and one counter electrode.

Other important research we can use :
Paper: Eco-Friendly Conformal and Self-Adhesive Electrochemical Sensors for Sweat Monitoring (2025)

- They developed flexible electrochemical sensors for sweat monitoring

They explain how they attached the alcohol detecting enzyme to the electrode
They used

- Alcohol oxidase : enzyme that helps convert alcohol (ethanol) into other chemicals through a chemical reaction
- Bovine serum albumin : protein extracted from cow blood that is commonly used to stabilize enzymes and other materials in lab experiments. In our project, this helps stabilize the alcohol oxidase enzyme when researchers attach it to an electrode

- Glutaraldehyde : chemical used to bind proteins together. Help attach and stabilize the alcohol oxidase enzyme on the electrode
- Nafion : synthetic polymer that researchers use to create thin, protective membranes on electrochemical sensors. It overall helps control which substances reach the electrode (there are many other substances in sweat, not just ethanol, so it kind of ‘filters’ what substances reach the electrode.

Paper also describes electrochemical characterization and testing in artificial human sweat - this is good for us because we can understand/learn how to construct the actual sensing surface rather than buying it (idk if well have time for this but good to check out)